

Narguess Alizade

Physics PhD candidate in theoretical/computational biophysics.

📍 Virginia Tech, USA

Education

Virginia Tech , Physics	Blacksburg, VA, USA Jan 2025 – present
Virginia Tech , Condensed Matter Physics	Blacksburg, VA, USA Jan 2023 – Jan 2025
Shahid Beheshti University (SBU) , Physics	Tehran, Iran Jan 2018 – Jan 2022
National Organization for Development of Exceptional Talents (Sampad) , Mathematics	Sari, Iran Jan 2012 – Jan 2018

Experience

Virginia Tech, Department of Physics, Center for Soft Matter and Biological Physics , Graduate Research Assistant Polyelectrolyte hydrogel diffusiophoresis. Supervised by Prof. C. N. Kaplan.	Blacksburg, VA Feb 2025 – present 1 year 8 months
Virginia Tech, Department of Physics, Center for Soft Matter and Biological Physics , Graduate Research Assistant Experimental evidence for the hydrogel diffusiophoresis mechanism. Supervised by Prof. C. N. Kaplan and Prof. J. R. Barone.	Blacksburg, VA Mar 2025 – present 1 year 7 months
Center for Complex Networks and Social Data Science, Shahid Beheshti University , Research Assistant Twitter sentiment analysis. Supervised by Prof. R. Jafari. • Conducted Twitter network analysis and performed sentiment classification using natural language processing, topic modeling, and deep learning.	Tehran, Iran July 2022 – Dec 2022 6 months
Institute for Research in Fundamental Sciences (IPM) , Research Assistant Breast cancer diagnosis. Supervised by Prof. S. Shojaeilangari. • Analyzed breast thermal images using deep learning for classification and prediction of healthy versus cancer cases.	Tehran, Iran May 2022 – Sept 2022 5 months
Tarbiat Modares University , Research Assistant Cell mechanical features and metastasis. Supervised by Prof. A. Allahverdi. • Analyzed fluorescent and confocal cell images to extract mechanical features and identify metastatic cells from extracted mechanical properties.	Tehran, Iran July 2021 – Dec 2021 6 months
Department of Physics, Virginia Tech , Teaching Assistant - PHYS 4455, Intro to Quantum Mechanics	Blacksburg, VA Jan 2026 – May 2026 5 months
Department of Physics, Virginia Tech , Teaching Assistant - PHYS 2215, 2216, 2305, 2306, General Physics	Blacksburg, VA Aug 2023 – May 2025 1 year 10 months

Department of Physics, Shahid Beheshti University , Teaching Assistant - Thermodynamics and Statistical Mechanics II	Tehran, Iran Sept 2022 – Jan 2023 5 months
Department of Physics, Shahid Beheshti University , Teaching Assistant - Solid State Physics I	Tehran, Iran Feb 2022 – June 2022 5 months
Department of Physics, Shahid Beheshti University , Teaching Assistant - Coding Sessions, School for Statistical Analysis of Cosmic Fields	Tehran, Iran Oct 2021 – Oct 2021 1 month
Department of Physics, Shahid Beheshti University , Teaching Assistant - General Physics I and II	Tehran, Iran Sept 2020 – Jan 2021 5 months
High School , High School Physics and Mathematics Teacher	Jan 2018 – Jan 2022 4 years 1 month

Volunteer

American Physical Society , APS Student Ambassador	Jan 2026 – Jan 2027
American Physical Society, Division of Biological Physics , APS-DBIO Student Committee Member	Jan 2026 – Jan 2028
Virginia Tech , Co-organizer, Soft Matter and Biological Physics (SMBP) Journal Club	Jan 2025 – present
Physics Department Graduate Committee, Virginia Tech , Graduate Student Representative	Jan 2026 – Jan 2028
Complex Networks Workshop , Executive Committee Member	May 2022 – May 2022
Two Days With Particle Physics Workshop , Executive Committee Member	Sept 2021 – Nov 2021
Scientific Association of Physics Students, Shahid Beheshti University , Secretary	Jan 2019 – Jan 2021
Physics Café, Shahid Beheshti University , Co-originator and organizer Co-originated and organized a weekly scientific gathering in the physics department for students and professors.	Jan 2018 – Jan 2021
Physics Day, Shahid Beheshti University , Organizing team Served on the organizing team for the department's annual outreach program.	Jan 2018 – Jan 2019

Awards

James R. Dunn Award Department of Physics, Virginia Tech	Apr 2026
Center for Soft Matter and Biological Physics Travel Award CSMBP, Virginia Tech	Sept 2025
Virginia Polytechnic Student Chapter Scholarship Award American Chemical Society, Rubber Band Division	Mar 2025
1st prize in the annual departmental poster session Department of Physics, Virginia Tech	May 2024

Ranked in the top 0.2% nationwide university entrance examination

Jan 2018

Ranked 448 among 300,000 participants

4th-place winner in the Khwarizmi Festival, Mechanics Division

Jan 2017

Young Scholars Club

Publications

Breast Cancer Automatic Diagnosis from Thermal Images Using Convolutional Neural Networks

Narguess Alizade, Seyedeh Samaneh Shojaeilangari, Mohammad Irandoost, Mohammad Firouzmand

A New Approach to the Equations of Free Fall Motion

Narguess Alizade, Aria Bozorgmehr

Skills

Programming & Computer

Machine Learning

LaTeX

FEniCS

Gephi

Mathematica

Languages

Interests

Research Interests

Certificates

UMass Summer School on Soft Solids and Complex Fluids

June 2026

APS Division of Fluid Dynamics (DFD) Annual Meeting

Nov 2025

Predictive Modeling in Biology and Medicine Conference

Nov 2023

The CODATA-RDA Research Data Science Summer School

July 2022

Computation in Complex Systems

Apr 2022

Finding Hidden Messages in DNA: Bioinformatics I

Feb 2022

Iranian Conference on Bioinformatics

Feb 2022

ICTP-ICTS Winter School on Quantitative Systems Biology

Dec 2021

The Annual Physics Conference of Iran

Oct 2021

Three-Day School of Introduction to CERN Activities

Sept 2021

Sociophysics School

Sept 2021

Spring School for Advanced Python and Machine Learning

Apr 2021

Introduction to Biological Evolution Summer School

July 2020